Orthonormal Basis Functions for Magnetic Localization of Submarine Telecommunications Cables: Closed-Form Position, Bearing and Standoff From a Single Crossing

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Abstract

Write a concise summary (150–250 words): context, problem, proposed method, main results, and significance.

Keywords: keyword one; keyword two; keyword three; keyword four

Nomenclature

Every symbol is defined again where it first appears, and this list gathers them with their units and the values used in the simulations. Vectors are bold. Horizontal positions are written (East, North) in metres, and three-dimensional vectors are in the local North–East–Down (NED) frame unless stated otherwise.

Symbol	Meaning	Unit	Value or range
Geometry and cable field			
I	DC feed current of the cable	A	1
$\mu_0/2\pi$	vacuum permeability over 2π	$\mathrm{nT\,m\,A^{-1}}$	200
d	standoff, vertical distance from the sensors to the cable	m	2–4
z_b, h_w	burial depth below the seabed, water depth	m	–
v	speed over ground	$\mathrm{m\,s^{-1}}$	1.0
θ	crossing angle (incidence) between the track and the cable	°	30–90
b	baseline between the two sensors (UNO port, DUO starboard)	m	1
A	snake amplitude, lateral excursion on each side of the cable	m	6
L, $K = L/d$	pass half-length, and the same in CPA distances	m, –	$K = 1.5$–3 on a snake leg (2)
$\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$, $\hat{\boldsymbol{z}}$	CPA frame axes (normal to the cable, along the cable, down)	–	–

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Symbol	Meaning	Unit	Value or range
x	horizontal coordinate along the cable normal, origin at the CPA	m	–
t_c	crossing time of a sensor	s	–
$\tau = x/d$	normalised coordinate, in CPA distances	–	(1)
$T_s = d/(v \sin\theta)$	duration of one CPA distance (template scale)	s	2.6–5.2 at $\theta = 50°$
$K_p = \mu_0 I/(2\pi d)$	peak cable field at the CPA	nT	100–50 (3)
$K' = \sqrt{\pi/2}\, K_p$	cable field scaled to the OBF coordinates	nT	–
$\boldsymbol{B}_s = (B_n, B_y, B_z)$	cable field in the CPA frame (normal, along the cable, down)	nT	(3)
ψ	angle from magnetic north to the cable normal	°	0–360
Geomagnetic field and channels			
$\boldsymbol{T}$, $\lvert\boldsymbol{T}\rvert$	geomagnetic field and its norm	nT	47 994 at Brest
$\hat{\boldsymbol{T}}$	unit vector of the geomagnetic field	–	–
δ	geomagnetic inclination (dip)	°	63.7 at Brest
S	total-field anomaly, $\lvert\boldsymbol{T} + \boldsymbol{B}_s\rvert - \lvert\boldsymbol{T}\rvert$	nT	–
Φ_1, Φ_2	cable orthonormal basis functions (even, odd)	–	(4)
$\mathcal{S}$	plane spanned by Φ_1 and Φ_2	–	–
$\mathcal{H}$	Hilbert transform	–	–
$\Psi = \Phi_1 + i\Phi_2$	analytic template	–	(7)
A_1, A_2	coordinates of the total-field anomaly on Φ_1 and Φ_2	nT	(5)
a, β	amplitude and phase of a channel in $\mathcal{S}$	nT, °	β: 64–116 for the total field at Brest (6)
$h = b\cos\theta/d$	sensor separation in the normalised coordinate	–	–
Γ_1, Γ_2	gradiometer orthonormal basis functions (odd, even)	–	(9)
Δs	gradiometer channel, difference DUO minus UNO	nT	(9)
Measurement model			
$\boldsymbol{F}_b$	field vector measured by a body-mounted sensor	nT	(10)
$\mathbf{E}_s$, $\boldsymbol{h}_s$	residual soft-iron matrix and hard-iron offset after calibration	–, nT	–
$\mathbf{R}^{nb}$	rotation matrix between the NED and body frames (roll, pitch, yaw χ)	–	–
χ	yaw of the vehicle	°	–
$\boldsymbol{B}_{ext}$	slow external field that is not the cable anomaly	nT	–
$\boldsymbol{n}_s$	sensor noise	nT	–

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Symbol	Meaning	Unit	Value or range
f_s	processing sample rate	Hz	50
Detection and truncation			
$m(\tau)$, m_k	recorded channel and its k-th sample	nT	–
t	lag of the matched filter, in CPA distances	–	–
$\boldsymbol{y}(t) = (y_1, y_2)$	matched-filter outputs at lag t	nT	(11)
$c_{ij}(t)$	correlation of Φ_i with Φ_j shifted by t	–	(13)
$\mathbf{C}(t)$	matrix of the correlations c_{ij}	–	–
$\Lambda(t)$	detection statistic	–	(15)
σ_n	noise standard deviation per sample	nT	–
η	detection threshold	–	calibrated empirically
$\mathcal{H}_0$, $\mathcal{H}_1$	hypotheses "no cable" and "cable"	–	–
P_{fa}	false-alarm probability	–	$\exp(-\eta/2)$ at one lag
$\mathbf{1}$	indicator function	–	–
c_{ij}^K, $\mathbf{C}^K$	correlations and their matrix under truncation to K	–	(40)
$E_i(K)$	energy of Φ_i inside $\lvert\tau\rvert \le K$	–	(17)
$I(K)$	integral of $\Phi_1\Phi_2'$ over $\lvert\tau\rvert \le K$	–	tends to $1/2$
D_1, D_2, X	terms of the truncated matched-filter energy	–	(41)
$\hat{\tau}_{\text{EP}}$	energy-peak displacement, in CPA distances	–	$4\sin 2\beta/(\pi K)$ (18)
k_0	coarse extremum of the first parity pass	samples	–
w, n_f	half-width of the symmetric template and of the fine search	samples	(19)
n	number of samples in a leg	–	–
m_z	vertical component of the cable field	nT	–
$\star$	correlation operator	–	–
Bearing and standoff			
$\boldsymbol{p}_k$	GNSS position (East, North) at sample k	m	–
$\boldsymbol{p}_{UNO,k}$, $\boldsymbol{p}_{DUO,k}$	sensor positions	m	(20)
$\hat{\boldsymbol{\ell}}(\chi)$	starboard unit vector	–	–
$\boldsymbol{p}_c$	crossing point, midpoint of the two fixes	m	–
φ	cable bearing, clockwise from north, modulo 180°	°	–
ε_1, ε_2	along-track errors of the UNO and DUO fixes	m	–
Q_j, Q_j^G	matched-filter outputs of the component j (East or North) of the field and of the gradiometer	nT	(24)
$\tilde{\Phi}_{1,s}$	unit-norm zero-mean template Φ_1 centred on the fix of sensor s	–	–
G_E, G_N	East and North components of the horizontal gradiometer vector	nT	–

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Symbol	Meaning	Unit	Value or range
w_i, σ_i	fusion weight and standard deviation of bearing estimator i	–, $^\circ$	$w_i = \sigma_i^{-2}$
$\hat{\boldsymbol{n}}$, $\hat{\boldsymbol{u}}$	unit vectors normal to and along the cable	–	–
x_k	coordinate of sample k along the estimated cable normal	m	(27)
$\hat{A}$	Φ_2 coefficient of the least-squares fit of a channel	nT	–
g	channel factor of AMP	–	1 for B_z, $\sin\delta$ for S
d_0, d_j	prior standoff and standoff at iteration j of AMP	m	$d_0 = 3$
$\lambda = d_j/d$	ratio of the template scale to the true standoff	–	–
κ	error ratio of the AMP iteration	–	1/2 on a long pass
$\epsilon = \Delta\varphi \cot\theta$	relative rescaling of x caused by a bearing error	–	–
Experiment			
SNR	ratio of the noise power to the total-field power at the CPA for $d = 1\,\mathrm{m}$ and $K \to \infty$	dB	–

Acronym	Meaning
CPA	closest point of approach
OBF	orthonormal basis functions
GLRT	generalised likelihood ratio test
SNR	signal-to-noise ratio
NED	North–East–Down frame
AHRS	attitude and heading reference system
GNSS	global navigation satellite system
UNO, DUO	port and starboard sensors of the vehicle
EP	energy-peak localiser
ZC	parity channel of the vertical component
GZ	parity channel of the gradiometer vertical component
TDOD	bearing from the time difference of detections of the two sensors
RAT	bearing from the OBF ratio
GRAT	bearing from the gradiometer ratio
FUS	fusion of the bearing estimators
AMP	standoff from the amplitude with known current
TRI	standoff from the two-sensor triangulation

1 Introduction

Submarine telecommunications cables carry most intercontinental data traffic [1], and their inspection, burial assessment, repair and protection all start with knowing where they are. Because the cable is often buried, acoustic and optical methods lose it, and electromagnetic

sensing becomes the main modality that penetrates the sediments. These cables carry a DC current of about 1 A to power the repeaters along the way, with the sea as the return path, and they therefore produce a static line-current field. At a distance of 2 m to 4 m from the cable this field lies between 50 nT and 100 nT, three orders of magnitude below the geomagnetic field of about 48 000 nT [2].

In previous work we derived two orthonormal basis functions, called cable OBF, matched to an infinite line current [3]. The resulting generalized likelihood ratio test (GLRT) detects the cable with fewer noise dimensions and places the closest point of approach (CPA) more accurately than dipole and multipolar receivers, both in simulation and in a lake experiment.

Detection alone does not follow a cable. A tracker must turn each detection into a description of the cable, fuse these descriptions over successive passes, and decide where the next pass goes, all under the motion and calibration errors of a small submarine drone [4–6].

This paper does not present the full autonomous tracking loop and focuses instead on the observation stage. We translate the filter output of the first paper into observations of the cable that do not depend on the length of the pass, namely its location, bearing and standoff. By working in the latent space spanned by the cable OBF rather than on raw field measurements, where the data are less noisy and more consistent, the paper bridges two research communities. The first is geometry-based cable tracking in robotics, and the second is signal processing for low signal-to-noise ratio. Methods from robotics can thus operate on the latent space of the cable OBF.

The contributions of this paper are the following:

- an amplitude and phase description of every OBF channel,
- a localization whose noiseless displacement under truncation is zero,
- closed-form expressions that extend the observation to bearing and standoff.

We assume a submarine drone moving at constant depth and velocity that follows the cable by repeatedly crossing it in a zigzag pattern.

The rest of the paper is organized as follows. Section 2 presents the survey geometry and the signal model. Section 3 addresses detection and localization under truncation. Section 4 derives the bearing and standoff observations from a single crossing. **??** reports the experiments.

2 Geometry and signal model

Every channel used in this paper is, to first order, a combination of two fixed functions of one normalised coordinate. this section sets the geometry, derives that coordinate and lists what the platform actually measures.

2.1 Survey Geometry

The vehicle moves at speed $v = 1.0\,\mathrm{m\,s^{-1}}$ along straight legs that cross a straight cable at an incidence θ (Figure 1a). Two tri-axial fluxgates are mounted at vehicle's depth on the vehicle's lateral axis, UNO on the port side and DUO on the starboard side, separated by the baseline

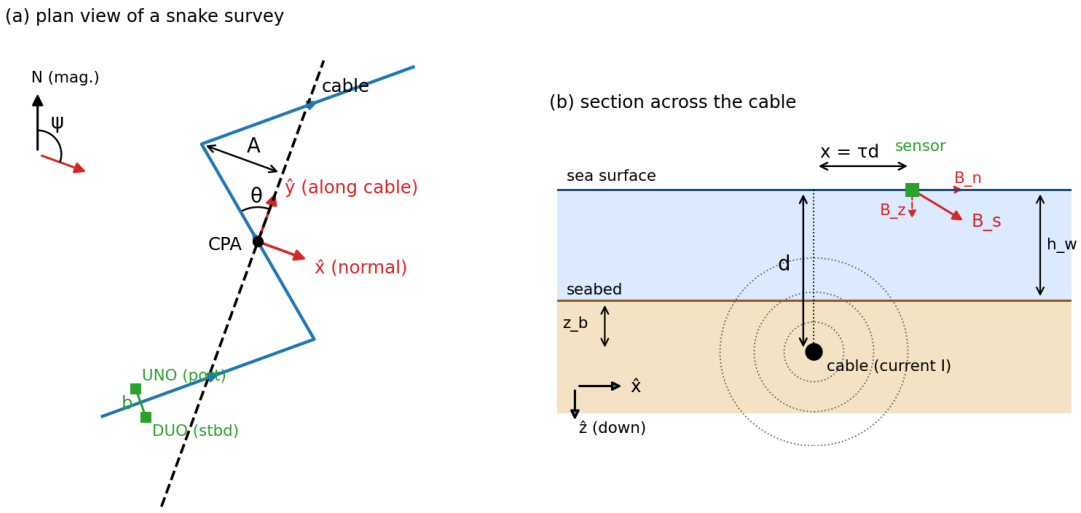


Figure 1: Survey geometry. (a) Plan view of a snake survey: cable (dashed), legs crossing it at incidence θ with amplitude A, CPA frame ($\hat{\boldsymbol{x}}$ normal, $\hat{\boldsymbol{y}}$ along the cable), the two sensors UNO and DUO separated by the baseline b, and the angle ψ from magnetic north to the cable normal. (b) Section across the cable: the sensor at the surface, the cable at standoff d below it (burial depth z_b below a seabed at water depth h_w), the normal coordinate $x = \tau d$, and the azimuthal cable field $\boldsymbol{B}_s$ with its normal and vertical components B_n and B_z. Change sea surface to vehicle depth

$b = 1\,\mathrm{m}$. For each sensor we use a local CPA frame: the origin is the point of its track directly above the cable, $\hat{\boldsymbol{x}}$ is the horizontal normal to the cable, $\hat{\boldsymbol{y}}$ points along the cable and $\hat{\boldsymbol{z}}$ points down (Figure 1b). The cable then lies at $x = 0$, $z = d$, where d is the standoff.

A sensor that crosses the cable at time t_c has horizontal distance $x = v\,(t - t_c)\sin\theta$ from the cable at time t. Every field quantity below depends on x only through the ratio x/d, so we use the normalised coordinate

$$\tau = \frac{x}{d} = \frac{v\,(t - t_c)\sin\theta}{d} = \frac{t - t_c}{T_s}, \qquad T_s = \frac{d}{v\sin\theta}. \tag{1}$$

τ counts CPA distances, and T_s is the time the sensor takes to move one CPA distance across the cable. T_s is the time scale of every matched template, and we call it the template scale. A leg that starts and ends at the lateral distance A from the cable covers $|x| \le A$, i.e. $|\tau| \le K$ with

$$K = \frac{L}{d}, \qquad L = A \text{ for a snake leg.} \tag{2}$$

L is the half-length of the pass measured across the cable. With the snake amplitude $A = 6\,\mathrm{m}$ used in closed loop and $d = 2\,\mathrm{m}$ to $4\,\mathrm{m}$, $K = 3$–1.5: a snake leg sees only a few CPA distances on each side of the crossing.

2.2 Cable Field and Cable Orthonormal Basis Functions

We follow the steps of [3]. The cable is modeled as an infinitely long straight conductor carrying a DC current I, with the return spread through the sea far from the sensor. Its field at the

sensor is azimuthal about the cable:

$$\mathbf{B}_s(\tau) = K_p \left(\frac{1}{1+\tau^2},\ 0,\ \frac{\tau}{1+\tau^2} \right)^T = (B_n, B_y, B_z)^T, \qquad K_p = \frac{\mu_0 I}{2\pi d}. \tag{3}$$

The components are given in the CPA frame: B_n is along the normal $\hat{x}$, $B_y = 0$ is along the cable, and B_z is downwards. K_p is the field magnitude at CPA, d is the standoff distance and $\tau = x/d$ is the normalized along-track position.

The infinite-line model holds when the cable is straight over several standoffs on each side of the crossing. The simulations instead use the exact field of a polyline made of 2 m segments.

Inserting (3) into the first-order term of the total-field anomaly gives a sum of two fixed functions of τ. Normalised to unit energy, they are the cable OBF of [3]:

$$\Phi_1(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1+\tau^2}, \qquad \Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{\tau}{1+\tau^2}. \tag{4}$$

Φ_1 is even and Φ_2 is odd. Both have unit energy over the real line, and they are orthogonal (Appendix A.2). The total-field anomaly is then

$$S(\tau) = A_1 \Phi_1(\tau) + A_2 \Phi_2(\tau), \qquad A_1 = \sqrt{\frac{\pi}{2}}\, K_p \cos\delta \cos\psi, \qquad A_2 = \sqrt{\frac{\pi}{2}}\, K_p \sin\delta. \tag{5}$$

The factor $\sqrt{\pi/2}$ undoes the normalisation of (4). A_1 and A_2 are the coordinates of S in the plane spanned by Φ_1 and Φ_2, and the matched filter of Section 3.1 estimates them. The expansion of the norm that justifies keeping only the first-order term, and the resulting phase, are given in Appendix A.1.

Any channel that is a fixed linear combination of the cable field can be written as one vector of $\mathcal{S} = \mathrm{span}\{\Phi_1, \Phi_2\}$:

$$s(\tau) = a \left(\cos\beta\, \Phi_1(\tau) + \sin\beta\, \Phi_2(\tau) \right), \qquad a \geq 0, \quad \beta \in (-180^\circ, 180^\circ]. \tag{6}$$

The amplitude a sets the signal level and the phase β sets the shape: $\beta = 0$ is the even bell Φ_1, $\beta = 90^\circ$ is the odd profile Φ_2, and intermediate phases are asymmetric (Figure 2c).

Hilbert relation. The observation underlying this paper is that Φ_2 is the Hilbert transform of Φ_1. The pair is therefore the real and imaginary parts of a single analytic template:

$$\Phi_2 = \mathcal{H}\{\Phi_1\}, \qquad \Psi(\tau) = \Phi_1(\tau) + i\,\Phi_2(\tau) = \sqrt{\frac{2}{\pi}} \frac{1}{1-i\tau}, \tag{7}$$

and the amplitude and phase of (6) are the modulus and argument of one complex coefficient:

$$s(\tau) = \Re\left\{ a\, e^{-i\beta}\, \Psi(\tau) \right\}. \tag{8}$$

This is the line-source instance of the classical property that the horizontal and vertical components of a 2D potential field form a Hilbert pair. Every channel the platform can form is thus one point $a\, e^{i\beta}$ of the complex plane, and the whole paper is an analysis of what each processing step does to that point.

Two properties of Ψ are used (Appendix A.3). Its spectrum is one-sided, so Φ_1 and Φ_2 have the same autocorrelation. The matched filter of Section 3.1 is then a quadrature receiver, whose output energy is the squared modulus of a single complex correlation and is therefore independent of β when the record is unbounded. And Ψ continues analytically off the real axis with a single pole at $\tau = -i$, which is what makes the coherence of a window with the span equal to one at the CPA and nowhere else (Appendix A.3). What truncation destroys is the first property, and the displacement it produces is proportional to $\sin 2\beta$.

In the implementation, however, Φ_1 and Φ_2 are used as two real templates and no estimator manipulates complex quantities. The analytic form is a bookkeeping device for the amplitude and the phase.

Gradiometer channels. The gradiometer is the difference DUO − UNO. The lateral axis makes the angle θ with the cable normal, so the two sensors sample the same channel at normal coordinates that differ by $b\cos\theta$, i.e. by $h = b\cos\theta/d$ in τ. To first order in that difference (a central difference),

$$\begin{aligned} \Delta s(\tau) &\simeq \frac{b\cos\theta}{d}\,\frac{\mathrm{d}s}{\mathrm{d}\tau} = \frac{b\cos\theta}{\sqrt{2}\,d}\,a\,(\cos\beta\,\Gamma_1(\tau) + \sin\beta\,\Gamma_2(\tau))\,, \\ \Gamma_1 = \sqrt{2}\,\Phi_1' &= -\frac{4}{\sqrt{\pi}}\frac{\tau}{(1+\tau^2)^2}, \qquad \Gamma_2 = \sqrt{2}\,\Phi_2' = \frac{2}{\sqrt{\pi}}\frac{1-\tau^2}{(1+\tau^2)^2}. \end{aligned} \tag{9}$$

The factor $\sqrt{2}$ makes Γ_1 and Γ_2 unit-energy: with the integrals of Appendix A.2 and $(1-\tau^2)^2 = (1+\tau^2)^2 - 4\tau^2$,

$$\int \Phi_1'^2 = \frac{2}{\pi}\int \frac{4\tau^2}{(1+\tau^2)^4} = \frac{1}{2}, \qquad \int \Phi_2'^2 = \frac{2}{\pi}\left(\int \frac{\mathrm{d}\tau}{(1+\tau^2)^2} - 4\int \frac{\tau^2}{(1+\tau^2)^4}\right) = \frac{1}{2}.$$

Φ_1' is odd and Φ_2' is even, so Γ_1 and Γ_2 are orthogonal by parity (Figure 2b): the vertical gradiometer ΔB_z follows the even profile Γ_2 and the horizontal gradiometer follows the odd profile Γ_1. Since $B_y = 0$ everywhere, the horizontal cable field points along $\hat{x}$ at both sensors, and so does their difference: the horizontal gradiometer vector lies along the cable normal. The first-order model (9) follows from $s(\tau + h/2) - s(\tau - h/2) = h\,s'(\tau) + (h^3/24)\,s'''(\tau) + O(h^5)$, so its relative error is of order $h^2/24$ times the ratio of the third to the first derivative.

The gradiometer amplitude is reduced by $b|\cos\theta|/(\sqrt{2}\,d)$, and it vanishes at perpendicular incidence.

The channels of this paper have the following amplitudes and phases (with $K' = \sqrt{\pi/2}\,K_p$):

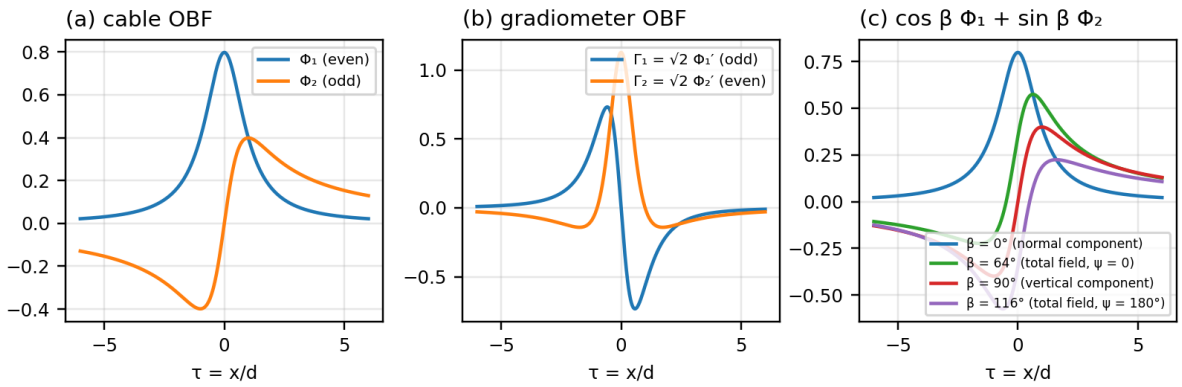


Figure 2: (a) Cable OBF: the even Φ_1 and the odd Φ_2 of (4). (b) Gradiometer OBF: the odd Γ_1 and the even Γ_2 of (9). (c) Channels of (6) for $\beta = 0°$ (normal component), 90° (vertical component), and the limits 64° and 116° of the total-field phase at Brest.

Table 2: Amplitude and phase of each channel in the basis $\{\Phi_1, \Phi_2\}$.

Channel	Amplitude a	Phase β
Normal component B_n	K'	0
Vertical component B_z	K'	90° (−90° for the opposite current)
North and East components, $B_n \cos\psi$ and $B_n \sin\psi$	$K'\lvert\cos\psi\rvert$, $K'\lvert\sin\psi\rvert$	0 or 180°
Total field S	$K'\sqrt{\cos^2\delta\,\cos^2\psi + \sin^2\delta}$	$\tan\beta = \tan\delta/\cos\psi$
Gradiometer, horizontal normal ΔB_n	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	0 on the gradiometer basis (odd profile Γ_1)
Gradiometer, vertical ΔB_z	$\dfrac{K'\,b\,\lvert\cos\theta\rvert}{\sqrt{2}\,d}$	90° on the gradiometer basis (even profile Γ_2)

The vector components have phases fixed by physics. The total-field phase follows from (5), $\tan\beta = A_2/A_1 = \tan\delta/\cos\psi$ (Appendix A.1). Since $\lvert\cos\psi\rvert \le 1$, we have $\lvert\tan\beta\rvert \ge \tan\delta$, so β lies between δ and $180° - \delta$: between 63.7° and 116.3° at Brest, for every cable bearing. The total field is therefore always closer to the odd profile than to the even one. At $\psi = 45°$, for example, $\beta = 70.7°$.

2.3 What the Platform Measures

Each body-mounted sensor records the body-frame vector

$$\boldsymbol{F}_b = (\mathbf{I} + \mathbf{E}_s)\,\mathbf{R}^{nb}(\text{roll}, \text{pitch}, \chi)\,(\boldsymbol{T} + \boldsymbol{B}_s + \boldsymbol{B}_{ext}) + \boldsymbol{h}_s + \boldsymbol{n}_s. \tag{10}$$

Here $\boldsymbol{B}_{ext}$ is an external magnetic field that is not the cable anomaly, and $\mathbf{E}_s$ and $\boldsymbol{h}_s$ are the soft-iron matrix and the hard-iron offset left after sensor calibration, which also absorb the static field of the platform that calibration did not account for. $\mathbf{R}^{nb}$ is the rotation matrix from the NED frame to the body frame, with χ the yaw of the vehicle, and applying it requires the attitude to be known. $\boldsymbol{n}_s$ is the sensor noise.

Three channels are formed from the recorded field.

- **Total field $|\boldsymbol{F}_b|$.** This channel is the norm of the measured components. Its main advantage is its independence from the attitude, so the errors of attitude estimation are absent.
- **Vector components in NED.** Contrary to [2], the processing is done in the NED frame, so the data are expressed in a frame that follows the vehicle but whose rotation is locked to north and east. Removing the regional field depends on the attitude estimate, which leaks an error from the regional field into the components that is comparable to the cable anomaly in this channel. This effect is tested in the later sections.
- **Gradiometer DUO − UNO.** $\boldsymbol{T}$ and $\boldsymbol{B}_{ext}$ are common to both sensors and cancel, so an attitude error then rotates only the difference vector. The attitude is still needed to orient the measurements in NED before the difference is taken.

The total field needs no attitude but mixes the two phases, so its shape changes with β. The vector components have a pure phase, so their shape does not vary with β, but they need the attitude. The gradiometer also has a pure phase and needs only the heading from the attitude, at the cost of an amplitude reduced by the factor $b|\cos\theta|/(\sqrt{2}\,d)$. This trade-off organises every comparison that follows.

3 Detection and localization under truncation

The length of the matched-filter projection onto $\mathcal{S}$ decides whether a cable is present, and its position decides where. With an unbounded record both are maximal at the CPA, but on a finite pass only quantities that do not depend on that length still place the crossing correctly. This section shows why, and gives the localisers that follow from it.

3.1 Matched Filter and Detection

Let $m(\tau)$ be one recorded channel over a leg, expressed in the normalised coordinate (1). The matched filter correlates it with the two basis functions centred at the lag t (in CPA distances):

$$y_i(t) = \int m(\tau)\,\Phi_i(\tau - t)\,\mathrm{d}\tau, \quad i = 1, 2, \qquad \mathbf{y}(t) = \big(y_1(t),\, y_2(t)\big)^\top \tag{11}$$

In the implementation the integral is a sum over samples m_k taken at $\tau_k = k/(f_s T_s)$, with templates sampled on the same grid and normalised to unit Euclidean norm. For a channel (6) and an unbounded record, (11) is linear in $(\cos\beta, \sin\beta)$:

$$\mathbf{y}(t) = a\,\mathbf{C}(t)^\top \begin{pmatrix}\cos\beta\\ \sin\beta\end{pmatrix}, \qquad \mathbf{C}(t) = \begin{pmatrix} c_{11} & c_{12}\\ c_{21} & c_{22}\end{pmatrix}, \qquad c_{ij}(t) = \int \Phi_i(\tau)\,\Phi_j(\tau - t)\,\mathrm{d}\tau. \tag{12}$$

$c_{ij}(t)$ is the correlation of Φ_i with Φ_j shifted by t. The four functions have closed forms (Appendix B.1):

$$c_{11}(t) = c_{22}(t) = \frac{4}{4 + t^2}, \qquad c_{21}(t) = -c_{12}(t) = \frac{2t}{4 + t^2}. \tag{13}$$

The identity $c_{21} = -c_{12}$ follows from parity alone (Φ_1 even, Φ_2 odd). The identity $c_{22} = c_{11}$ is the consequence of the Hilbert relation announced in Section 2.2: Φ_1 and Φ_2 have the same autocorrelation. Together they make $\mathbf{C}(t)$ a rotation matrix multiplied by the scalar $2/\sqrt{4+t^2}$, and a rotation does not change the length of the vector $(\cos\beta, \sin\beta)$. Hence

$$\|\mathbf{y}(t)\|^2 = y_1^2 + y_2^2 = \frac{4a^2}{4+t^2} \qquad \text{for every phase } \beta. \tag{14}$$

The energy is maximal at $t = 0$, the CPA, falls to half at $|t| = 2$, and does not depend on which channel is used. With an unbounded record the energy peak therefore locates the crossing for every channel and every geometry.

Detection. The detector thresholds the energy normalised by the noise variance:

$$\Lambda(t) = \frac{\|\mathbf{y}(t)\|^2}{\sigma_n^2} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \eta. \tag{15}$$

$\mathcal{H}_0$ is the hypothesis "no cable" and $\mathcal{H}_1$ "cable". σ_n is the noise standard deviation per sample and η the threshold. Under white Gaussian noise, (15) is the GLRT for a signal in $\mathcal{S}$ with unknown amplitudes [3, 7, 8]. With unit-norm, orthogonal templates, $\mathbf{y}$ is then Gaussian with covariance $\sigma_n^2\mathbf{I}$ under $\mathcal{H}_0$, so Λ follows a χ^2 law with 2 degrees of freedom and the false-alarm probability at one lag is $P_{\text{fa}} = \exp(-\eta/2)$ (Appendix B.2). In practice the threshold is calibrated empirically instead.

3.2 Truncation Breaks the Phase Invariance

A leg restricts the record to $|\tau| \le K$. Two truncation models must be distinguished:

$$\begin{aligned} \text{template truncation:} \quad & y_i(t) = \int m(\tau)\, \Phi_i(\tau - t)\, \mathbf{1}_{|\tau - t| \le K}\, \mathrm{d}\tau, \\ \text{record truncation:} \quad & y_i(t) = \int m(\tau)\, \mathbf{1}_{|\tau| \le K}\, \Phi_i(\tau - t)\, \mathrm{d}\tau. \end{aligned} \tag{16}$$

$\mathbf{1}$ is the indicator function. With template truncation the record is unbounded and the templates are clipped to K CPA distances about their centre. With record truncation the record is clipped to the leg and correlated with full templates. A survey leg is a record truncation, because the implementation correlates long templates ($|\tau| \le 30$) with whatever record exists.

In both models (12) still holds, with truncated correlations c_{ij}^K. Parity still gives $c_{12}^K(0) = c_{21}^K(0) = 0$, but the equal-autocorrelation identity fails. At $t = 0$ the two diagonal terms are the energies of Φ_1 and Φ_2 inside the pass:

$$\begin{gathered} E_1(K) = \frac{2}{\pi}\Big(\arctan K + \frac{K}{1+K^2}\Big), \qquad E_2(K) = \frac{2}{\pi}\Big(\arctan K - \frac{K}{1+K^2}\Big), \\ E_1 - E_2 = \frac{4K}{\pi(1+K^2)}. \end{gathered} \tag{17}$$

Φ_2 decays as $1/\tau$ while Φ_1 decays as $1/\tau^2$, so Φ_2 keeps more of its energy beyond the pass: $E_2 < E_1$. $\mathbf{C}^K(t)$ is no longer a scaled rotation, and the energy acquires a term proportional to $\sin 2\beta$ that is odd in t. Its slope at $t = 0$ moves the maximum. To first order in t (Appendix B.3) the energy peak sits at

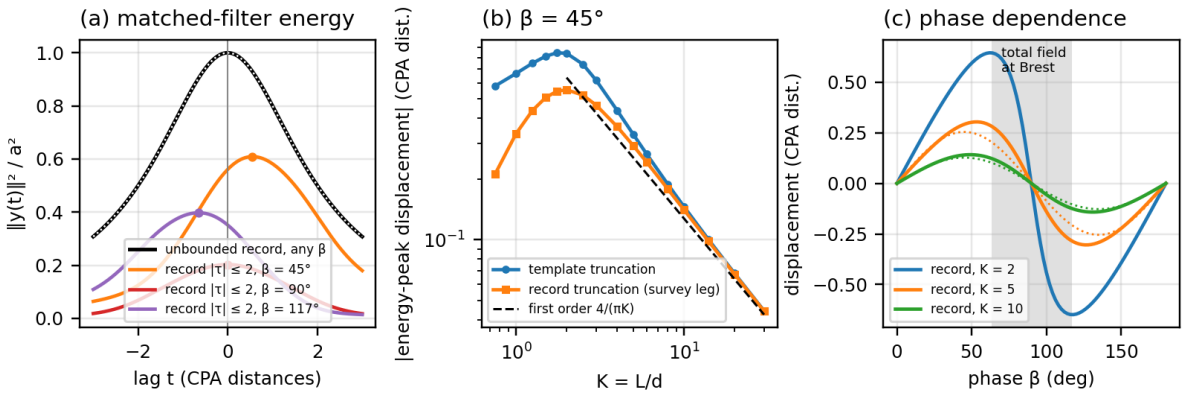


Figure 3: (a) Matched-filter energy $\|\mathbf{y}(t)\|^2/a^2$ for an unbounded record (the same curve $4/(4+t^2)$ for every β, black) and for a record truncated to $|\tau| \le 2$ at $\beta = 45°$, 90° and 117° (dots: maxima). (b) Energy-peak displacement at $\beta = 45°$ versus K for template and record truncation, with the first-order law $4/(\pi K)$. (c) Displacement versus β under record truncation for $K = 2$, 5 and 10, with the first-order law (dotted, $K = 5$ and 10). The shaded band is the total-field phase range at Brest.

$$\hat{\tau}_{\mathrm{EP}} \simeq \frac{4}{\pi K} \sin 2\beta, \qquad K \gg 1. \tag{18}$$

The displacement is driven by the energy imbalance $E_1 - E_2 \approx 4/(\pi K)$ between the truncated basis functions: it tilts the energy about the CPA, and the maximum moves towards the side the tilt favours, by an amount proportional to $\sin 2\beta$. Both truncation models share this law. The derivation, and its evaluation at finite K, are given in Appendix B.3.

Three properties follow. The displacement vanishes for the pure phases $\beta = 0°$ and 90°, i.e. for the horizontal and vertical components. It changes sign with β, so it cannot be averaged out over passes of unknown phase. And it decays only as $1/K$: halving it requires doubling the pass.

Equation (18) is a first-order result. The exact displacement at $\beta = 45°$ exceeds it by about 30% at $K = 5$ (0.33 against 0.25 CPA distances), 15% at $K = 10$, 7% at $K = 20$ and 4% at $K = 30$ (Figure 3b). For short passes the higher-order terms dominate. Under record truncation, the survey case, the largest displacement over all phases and $K \ge 0.75$ is 0.65 CPA distances, at $K = 2$ and $\beta \approx 117°$. Within the total-field phase range at Brest (64–116°) it is 0.65, 0.28 and 0.12 CPA distances at $K = 2$, 5 and 10 (Figure 3c). At $d = 2$–4 m, 0.65 CPA distances is 1.3–2.6 m. Under template truncation the energy becomes bimodal as β approaches 90° at $K \approx 2$ (Figure 4a), so its maximum jumps between two lobes about two CPA distances apart. Record truncation does not show this bimodality (Figure 4b).

The lawnmower lines of [3] operated at $K \gg 10$, where the displacement is small. A snake leg of amplitude 6 m over a standoff of 2–4 m operates at $K = 1.5$–3, where it is largest.

3.3 Parity Channels

A channel with a constant phase is not affected by truncation. The vertical component $a\,\Phi_2$ is odd about the CPA, and its correlation with the template Φ_2 is an even function of the lag centred at the CPA, under one constraint: the truncation has to be symmetric about the CPA. The extremum of this correlation then marks the crossing, and we call this parity localiser (or

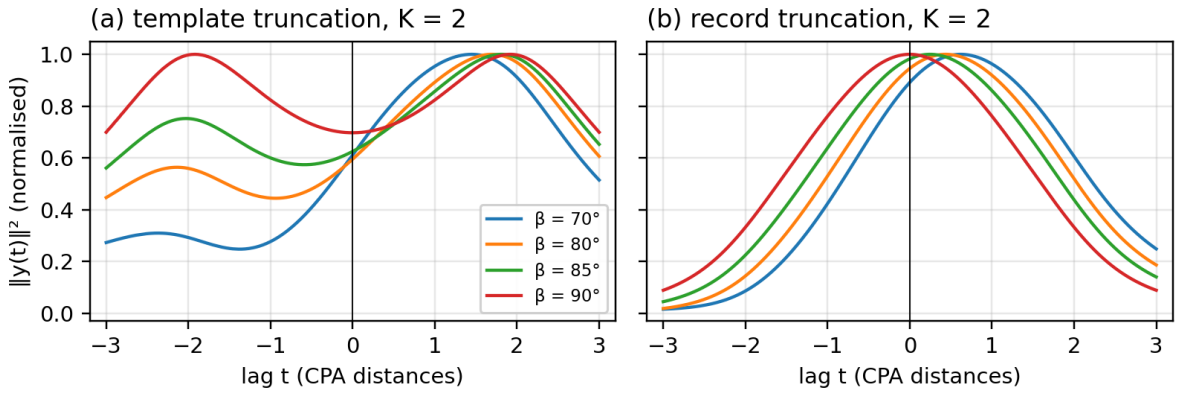


Figure 4: Normalised energy profiles at $K = 2$ for $\beta = 70$–$90°$. (a) Template truncation, which becomes bimodal as β approaches 90°. (b) Record truncation, which does not.

zero crossing) ZC.

On a snake leg the record of each sensor is not symmetric about its own crossing. The vehicle centre crosses the cable in the middle of the leg, but a sensor offset by $b/2$ along the lateral axis crosses it earlier or later. In addition, the leg is planned on the estimated cable, so the actual crossing is displaced from the middle of the leg by the error of that estimate, and this asymmetry cannot be compensated in advance.

We therefore localise in two passes.

1. A template of half-width 30 CPA distances gives the coarse extremum k_0 of $|m_z \star \Phi_2|$ within $\pm T_s$ of the detection, where m_z is the vertical component and $\star$ denotes correlation.

2. The template is truncated symmetrically about k_0 to the half-width

$$w = \min(k_0,\, n - 1 - k_0) - n_f, \qquad n_f = T_s f_s, \qquad \max(0.5\, T_s f_s,\, 2) \le w \le 30\, T_s f_s, \quad (19)$$

   in samples, so that it matches the truncation of the recorded signal.

Here n is the number of samples in the leg and n_f the half-width of the fine search. Every lag within $\pm n_f$ of k_0 then sees a complete record over the support of its template, which restores exact parity.

The gradiometer vertical channel ΔB_z follows Γ_2 (9), which is even about the crossing of the baseline centre. Its correlation with Γ_2 is again even in t, so the same two-pass search applies (GZ). GZ gives one fix at the vehicle position, whereas ZC gives one fix per sensor and the crossing point is the midpoint of the two fixes.

4 Extended Observation: Bearing and Standoff From One Crossing

A validated crossing gives a point on the cable, and the same records also give the cable bearing and the standoff. The point $\mathbf{p}_c$ is the midpoint of the two sensors' fixes, or the vehicle position

at the GZ fix. The positions of the two sensors are computed from the navigation solution:

$$\mathbf{p}_{UNO,k} = \mathbf{p}_k - \frac{b}{2}\,\hat{\boldsymbol{\ell}}(\chi_k), \qquad \mathbf{p}_{DUO,k} = \mathbf{p}_k + \frac{b}{2}\,\hat{\boldsymbol{\ell}}(\chi_k), \qquad \hat{\boldsymbol{\ell}}(\chi) = (\cos\chi, -\sin\chi). \tag{20}$$

Here $\mathbf{p}_k$ is the position (East, North) given by the global navigation satellite system (GNSS) at sample k, χ_k the yaw given by the attitude and heading reference system (AHRS) and $\hat{\boldsymbol{\ell}}$ the starboard unit vector.

The classical closed-form estimators of bearing and standoff read the field at individual samples, as a ratio of components or a peak value. At low signal-to-noise ratio (SNR) a single sample is noisy. TDOD, RAT, GRAT and AMP below therefore work on projections of the records onto the cable OBF $\{\Phi_1, \Phi_2\}$ or the gradiometer OBF $\{\Gamma_1, \Gamma_2\}$ of Section 2, which integrate the whole crossing. TRI is kept in its sample-wise form as the reference method of the cable-tracking literature. All of them are in closed form. Constants are treated as nuisance parameters throughout, through a constant column in each least-squares fit, as in the matched-subspace detector with interference [8]. Table 3 lists the estimators, what each one needs, and the raw-field method it replaces.

Table 3: Bearing and standoff estimators of one crossing. S is the total field, Attitude is what the estimator needs from the AHRS, and Form is a closed form, a same-shape ratio or a closed-form step iterated to a fixed point. AMP and TRI also use the bearing, through the normal coordinate (27) or the incidence θ.

Estimator	Channel	Attitude	Current	Form	Raw-field method
TDOD	ZC fixes of B_z	full	no	closed	line through two fixes [9]
RAT	B_E, B_N	full	no	ratio	heading deviation [4]
GRAT	G_E, G_N	heading	no	ratio	gradient direction [10]
FUS	TDOD, RAT, GRAT	as inputs	no	closed	–
AMP	B_z or S	full or none	yes	fixed point	level, known current [10]
TRI	B_n, B_z, two sensors	full	no	closed	two inclinations [4]

4.1 Bearing

Time difference of detection (TDOD). Both sensor fixes lie on the cable, so the bearing is the direction that joins them, as in crossing-based trackers [9]:

$$\hat{\varphi}_{TDOD} = \operatorname{atan2}\left(E_{DUO} - E_{UNO},\, N_{DUO} - N_{UNO}\right) \bmod 180^\circ. \tag{21}$$

(E, N) are the East and North coordinates of each fix in metres, and φ is measured clockwise from north. Because it is written in positions, TDOD absorbs speed variations. Along-track errors ε_1 and ε_2 of the UNO and DUO fixes give the bearing error

$$\Delta\hat{\varphi}_{TDOD} \simeq \frac{(\varepsilon_2 - \varepsilon_1)\sin^2\theta}{b}, \tag{22}$$

which grows with the standoff, since the fix errors scale with $d/\sin\theta$. The two fixes are the ZC fixes of B_z, so TDOD depends on the attitude. GZ gives a single fix, so TDOD does not apply to it.

Same-shape ratios. The two other bearings rest on one property. When two channels have the same waveform up to a constant factor, the ratio of their least-squares coefficients on any template equals that factor exactly, whatever the template scale, its position and the truncation of the record (Appendix C.1). The template only sets the SNR of the result, which is highest when the template matches the waveform.

OBF ratio (RAT). The vector channels are rotated to the North–East–Down (NED) frame with the AHRS attitude. In NED the horizontal anomaly vector is aligned with the cable normal at every sample and follows the even profile Φ_1 (Table 2). Its East and North components, $B_n \sin(\varphi + 90°)$ and $B_n \cos(\varphi + 90°)$, therefore have the same shape, and their ratio gives the normal. This is the heading-deviation relation of [4, 6], written on projections rather than samples:

$$\hat{\varphi}_{RAT} = \operatorname{atan2}(Q_E, Q_N) + 90° \bmod 180°, \tag{23}$$

where Q_j is the matched-filter output of B_j at the fix lag, summed over the two sensors,

$$Q_j = \sum_{s \in \{UNO, DUO\}} \langle B_{j,s}, \tilde{\Phi}_{1,s} \rangle, \qquad j \in \{E, N\}, \tag{24}$$

and $\tilde{\Phi}_{1,s}$ is the unit-norm template Φ_1 centred on the fix of sensor s and made zero-mean over its window, which rejects the offsets of the components (Appendix C.1). As a same-shape ratio, RAT is exact under truncation and for any template scale, but it needs the full attitude. On the asymmetric windows of a snake leg the slow attitude drift does not cancel and leaks into the bearing.

Gradiometer ratio (GRAT). The horizontal gradiometer vector (G_E, G_N), the horizontal part of $\mathbf{B}_{DUO} - \mathbf{B}_{UNO}$, also lies along the cable normal, with the odd profile Γ_1 (9). Its two components are again a same-shape pair, and Q^G_j is the matched-filter output of G_j with the template Γ_1 centred on the fix and made zero-mean over its window:

$$\hat{\varphi}_{GRAT} = \operatorname{atan2}(Q^G_E, Q^G_N) + 90° \bmod 180°. \tag{25}$$

This is the direction of the field gradient used with differential sensors [10, 11], written on projections. Common-mode fields cancel in the difference, tilt errors enter at second order, and only the heading is needed. The amplitude is proportional to $|\cos\theta|$ and vanishes at perpendicular incidence.

Fusion (FUS). Bearings are axial, defined modulo 180°, so they are fused by an inverse-variance mean of the doubled angle [12]:

$$\hat{\varphi}_{FUS} = \frac{1}{2} \operatorname{atan2}\left(\sum_i w_i \sin 2\hat{\varphi}_i, \sum_i w_i \cos 2\hat{\varphi}_i \right), \qquad w_i = \sigma_i^{-2}, \tag{26}$$

with $i \in \{\mathrm{TDOD}, \mathrm{RAT}, \mathrm{GRAT}\}$. The σ_i are trimmed RMS errors read from tables indexed by standoff and incidence, calibrated on simulated single crossings (Section 6, E1). The weights

assume independent errors, which holds only approximately since TDOD and RAT share the AHRS. FUS combines the three bearings when the attitude is trusted, and GRAT is used alone otherwise.

4.2 Standoff

The coordinate of each sample along the estimated cable normal is

$$x_k = (\mathbf{p}_{s,k} - \mathbf{p}_c) \cdot \hat{\mathbf{n}}(\hat{\varphi}), \qquad \hat{\mathbf{n}}(\varphi) = (\cos\varphi, -\sin\varphi), \tag{27}$$

where $\mathbf{p}_{s,k}$ is the position (20) of sensor s. A bearing error $\Delta\varphi$ rescales x by about $(1 - \Delta\varphi \cot\theta)$ (Appendix C.2).

Amplitude with known current (AMP). The feed current of a repeatered cable is regulated by the shore station and known to the operator [1], and cable-locating systems have long used it to convert a field level into a distance [10]. In the OBF the conversion uses a single coefficient, that of the channel on the odd function Φ_2. For the vertical component this coefficient is $A_z = \sqrt{\pi/2}\, K_p$, and for the total field it is $A_2 = \sqrt{\pi/2}\, K_p \sin\delta$ (5), which does not depend on the cable bearing. With $K_p = \mu_0 I/(2\pi d)$ the standoff follows from the coefficient $\hat{A}$ of either channel,

$$\hat{d} = \frac{\mu_0 I}{2\pi}\sqrt{\frac{\pi}{2}}\,\frac{g}{|\hat{A}|}, \qquad g = 1 \text{ for } B_z, \qquad g = \sin\delta \text{ for the total field.} \tag{28}$$

$\hat{A}$ is the Φ_2 coefficient of the least-squares fit of $[\Phi_1(x_k/d), \Phi_2(x_k/d), 1]$ to the channel on a window of $\pm K$ CPA distances around the fix, averaged over the two sensors. Because this coefficient depends on the template scale d, AMP applies (28) as a fixed-point iteration,

$$d_{j+1} = \frac{\mu_0 I}{2\pi}\sqrt{\frac{\pi}{2}}\,\frac{g}{|\hat{A}(d_j)|}, \tag{29}$$

started from the prior standoff d_0 and stopped when the change falls below 1 cm. The true standoff is a fixed point for every truncation of the record, because a template at the right scale fits the record exactly. Each step halves the error on a long pass and reduces it much faster on a snake leg, so a few iterations suffice and the result does not depend on the starting value (Appendix C.2). Each step is a linear least-squares fit, so AMP needs no scan. With a trusted attitude AMP can use B_z. Otherwise it uses the total field, which needs no attitude, and together with GZ and GRAT the whole observation then needs only the heading. The bearing enters only through the template scale, and on a snake leg its effect is a small fraction of its effect on a fit of the profile shape (Appendix C.2).

The price of AMP is a direct sensitivity to the assumed current. The effective magnetic current can differ from the shore-station current, for instance through return currents along the armour near shore ends, and a current error of 10% biases $\hat{d}$ by about 10%. Heave modulates the amplitude in the same way.

Two-sensor triangulation (TRI). When the cable lies between the sensors ($x_{UNO}x_{DUO} < 0$), each sensor's inclination gives $|B_{z,i}|/|B_{n,i}| = |x_i|/d$, and $|x_{UNO}| + |x_{DUO}| = b|\cos\theta|$ [4], hence

$$\hat{d}_{TRI} = \frac{b|\cos\theta|}{|B_{z,UNO}|/|B_{n,UNO}| + |B_{z,DUO}|/|B_{n,DUO}|}, \tag{30}$$

taken as the median over the samples where the cable lies between the sensors. TRI is the reference two-sensor method of the cable-tracking literature [4–6], and it needs no current. It measures d against the lateral separation of the sensors, $b|\cos\theta|$, which is several times smaller than the standoff here, so small errors on the inclinations become large errors on d. The trivial baseline, the prior mean of 3 m, is reported alongside.

4.3 Observability

With position fixes only, one crossing constrains the cable to an infinite set of lines through a point. Two crossings are then needed for the bearing, and the standoff is never observed [9]. The bearing observation makes the horizontal line observable from a single crossing, and the standoff observation adds the depth of the cable below the sensors. Their practical value depends on their accuracy compared with the variability of the cable along the route, which Section 7 quantifies.

5 Experiment

to validate the observation and localisation methods, we set to conduct two experiments in simulation, where we have a fine control over the parameters and a precise ground truth.

the experiments are a motecarlos on the signal with simple GWAN to test the accuracy of every method, the second test is a long cable obsticle course where real noise model is used to disturb the signal it self, which will allow us to suggest a method based on the platform noise scenario.

5.1 Montecarlo

five parameters have been selected to study every method: beta the anomaly phase in $\mathcal{S}$, d the standoff, theta the crossing angle, K truncation in CPA dist, SNR at 1m.

where the SNR here is the ratio in db between the power of GWAN and the total field at CPA when d=1m and k tends to infinity.

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A Total Field and Cable Basis Functions

A.1 Expansion of the Total Field and Its Phase

A scalar magnetometer, or the norm of a vector magnetometer, measures $|\mathbf{T} + \mathbf{B}_s|$, where $\mathbf{T}$ is the geomagnetic field. With $u = 2\,\mathbf{B}_s \cdot \hat{\mathbf{T}}/|\mathbf{T}| + |\mathbf{B}_s|^2/|\mathbf{T}|^2$ we have $|\mathbf{T} + \mathbf{B}_s| = |\mathbf{T}|\sqrt{1+u} =$

$|\mathbf{T}|(1+u/2-u^2/8+O(u^3))$, so, keeping terms up to second order in $|\mathbf{B}_s|/|\mathbf{T}|$,

$$S=|\mathbf{T}+\mathbf{B}_s|-|\mathbf{T}|=\mathbf{B}_s\cdot\hat{\mathbf{T}}+\frac{|\mathbf{B}_s|^2-(\mathbf{B}_s\cdot\hat{\mathbf{T}})^2}{2|\mathbf{T}|}+O\!\left(\frac{|\mathbf{B}_s|^3}{|\mathbf{T}|^2}\right),\qquad \hat{\mathbf{T}}=\frac{\mathbf{T}}{|\mathbf{T}|}. \tag{31}$$

The second-order terms equal $|\mathbf{B}_{s\perp}|^2/(2|\mathbf{T}|)$, where $\mathbf{B}_{s\perp}$ is the part of $\mathbf{B}_s$ perpendicular to $\mathbf{T}$. They are at most $K_p^2/(2|\mathbf{T}|)=100^2/(2\times 47\,994)=0.10$ nT at $d=2$ m, i.e. 0.1% of the anomaly. Only the first term is kept in the analysis, and all simulations use the exact norm.

The horizontal part of $\hat{\mathbf{T}}$ has magnitude $\cos\delta$ and points to magnetic north, and $\hat{x}$ makes the angle ψ with magnetic north, so $\hat{\mathbf{T}}\cdot\hat{x}=\cos\delta\cos\psi$ and $\hat{\mathbf{T}}\cdot\hat{z}=\sin\delta$ ($\hat{\mathbf{T}}\cdot\hat{y}$ is irrelevant because $B_y=0$). With (3), the first-order term is

$$\mathbf{B}_s\cdot\hat{\mathbf{T}}=K_p\left(\frac{\cos\delta\cos\psi}{1+\tau^2}+\frac{\sin\delta\,\tau}{1+\tau^2}\right)=\sqrt{\frac{\pi}{2}}\,K_p\big(\cos\delta\cos\psi\,\Phi_1+\sin\delta\,\Phi_2\big), \tag{32}$$

which is (5), since $1/(1+\tau^2)=\sqrt{\pi/2}\,\Phi_1$ and $\tau/(1+\tau^2)=\sqrt{\pi/2}\,\Phi_2$. Its amplitude and phase in the sense of (6) are

$$a=\sqrt{A_1^2+A_2^2}=\sqrt{\frac{\pi}{2}}\,K_p\sqrt{\cos^2\delta\cos^2\psi+\sin^2\delta},\qquad \tan\beta=\frac{A_2}{A_1}=\frac{\tan\delta}{\cos\psi}, \tag{33}$$

with β in the quadrant of (A_1,A_2). Because $|\cos\psi|\le 1$, $|\tan\beta|\ge\tan\delta$, hence $\delta\le|\beta|\le 180^\circ-\delta$. At Brest $\tan\delta=43\,023/21\,271=2.023$ ($\delta=63.7^\circ$). For $\psi=45^\circ$, $\tan\beta=2.023/0.707=2.86$, so $\beta=70.7^\circ$ and $a=0.95\sqrt{\pi/2}\,K_p$.

A.2 Standard Integrals and Orthonormality

The integrals used throughout are

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}\tau}{(1+\tau^2)^2}=\frac{\pi}{2},\qquad \int_{-\infty}^{\infty}\frac{\mathrm{d}\tau}{(1+\tau^2)^3}=\frac{3\pi}{8},\qquad \int_{-\infty}^{\infty}\frac{\tau^2\,\mathrm{d}\tau}{(1+\tau^2)^4}=\frac{\pi}{16}. \tag{34}$$

Thus $\int\Phi_1^2=(2/\pi)(\pi/2)=1$, and since $\tau^2/(1+\tau^2)^2=1/(1+\tau^2)-1/(1+\tau^2)^2$, $\int\Phi_2^2=(2/\pi)(\pi-\pi/2)=1$. The product $\Phi_1\Phi_2$ is odd, so $\int\Phi_1\Phi_2=0$: the pair (4) is orthonormal over the real line. The last two integrals of (34) give the unit energy of the gradiometer basis (9).

A.3 The Analytic Template: Hilbert Pair, Spectrum and Poles

With the Fourier convention $\hat{f}(\omega)=\int f(\tau)\,e^{-i\omega\tau}\,\mathrm{d}\tau$, the transform of $1/(1+\tau^2)$ is $\pi e^{-|\omega|}$, so

$$\hat{\Phi}_1(\omega)=\sqrt{2\pi}\,e^{-|\omega|},\qquad \hat{\Phi}_2(\omega)=-i\,\mathrm{sgn}(\omega)\,\hat{\Phi}_1(\omega), \tag{35}$$

the second because the Hilbert transform multiplies a spectrum by $-i\,\mathrm{sgn}\,\omega$, and the inverse transform of $-i\pi\,\mathrm{sgn}(\omega)\,e^{-|\omega|}$ is $\tau/(1+\tau^2)$, hence $\Phi_2=\mathcal{H}\{\Phi_1\}$. Two consequences follow.

First, $|\hat{\Phi}_2|=|\hat{\Phi}_1|$, so by the Wiener–Khinchin theorem [13–15] Φ_1 and Φ_2 have the same autocorrelation function. Equivalently, the analytic template $\Psi=\Phi_1+i\Phi_2$ of (7) has the

one-sided spectrum

$$\hat{\Psi}(\omega) = (1+\operatorname{sgn}\omega)\hat{\Phi}_1(\omega) = \begin{cases} 2\sqrt{2\pi}\, e^{-\omega}, & \omega > 0, \\ 0, & \omega < 0, \end{cases} \qquad \int \Psi(\tau)\, \Psi^*(\tau - t)\, \mathrm{d}\tau = \frac{4}{2 - it}. \tag{36}$$

Correlating a channel with Ψ is therefore a quadrature receiver, and the modulus of (36), $4/\sqrt{4+t^2}$, carries no dependence on the phase β of (8): this is the phase invariance exploited in Section 3.1, and what record truncation destroys.

Second, $\Psi(\tau) = \sqrt{2/\pi}/(1 - i\tau)$ continues analytically off the real axis as a rational function with the single pole $\tau = -i$. A shifted template $\Psi(\tau - t)$ has its pole at $t - i$, and two rational functions that agree on an interval agree everywhere, so their poles must coincide: for real t this forces $t = 0$. Hence a window of the record lies in the span of the shifted pair only at $t = 0$: the subspace coherence of a window with $\operatorname{span}\{\Phi_1, \Phi_2\}$ reaches one at the CPA and nowhere else, and a wrong template scale λ, which moves the pole to $t - i\lambda$, prevents it from reaching one at all.

B Matched Filter and Truncation

B.1 Closed-Form Correlations

By Parseval's theorem [15], with the Fourier convention of Appendix A.3,

$$c_{ij}(t) = \int \Phi_i(\tau)\, \Phi_j(\tau - t)\, \mathrm{d}\tau = \frac{1}{2\pi} \int \hat{\Phi}_i(\omega)\, \hat{\Phi}_j^*(\omega)\, e^{i\omega t}\, \mathrm{d}\omega. \tag{37}$$

From (35), $\hat{\Phi}_1\hat{\Phi}_1^* = \hat{\Phi}_2\hat{\Phi}_2^* = 2\pi e^{-2|\omega|}$ and $\hat{\Phi}_1\hat{\Phi}_2^* = 2\pi\, i \operatorname{sgn}(\omega)\, e^{-2|\omega|}$. With the standard integrals $\int e^{-a|\omega|}\cos(\omega t)\, \mathrm{d}\omega = 2a/(a^2 + t^2)$ and $\int_0^\infty e^{-a\omega}\sin(\omega t)\, \mathrm{d}\omega = t/(a^2 + t^2)$ at $a = 2$,

$$\begin{aligned} c_{11} = c_{22} &= \int e^{-2|\omega|}\, e^{i\omega t}\, \mathrm{d}\omega = \frac{4}{4 + t^2}, \\ c_{12} &= \int i \operatorname{sgn}(\omega)\, e^{-2|\omega|}\, e^{i\omega t}\, \mathrm{d}\omega = -2\int_0^\infty e^{-2\omega}\sin(\omega t)\, \mathrm{d}\omega = -\frac{2t}{4 + t^2}. \end{aligned} \tag{38}$$

Substituting $\tau \to -\tau$ and using the parities of Φ_1 and Φ_2 gives $c_{21}(t) = -c_{12}(t) = 2t/(4 + t^2)$. With $\cos\vartheta = 2/\sqrt{4+t^2}$ and $\sin\vartheta = t/\sqrt{4+t^2}$,

$$\mathbf{C}(t) = \frac{2}{\sqrt{4+t^2}} \begin{pmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{pmatrix}, \tag{39}$$

a rotation scaled by $2/\sqrt{4+t^2}$. Hence $\|\mathbf{C}(t)^\top(\cos\beta, \sin\beta)^\top\| = 2/\sqrt{4+t^2}$ for every β, which is (14).

B.2 Detection Statistic Under the Null Hypothesis

Under $\mathcal{H}_0$ the record is white Gaussian noise of variance σ_n^2 per sample. With g_i the sampled, unit-norm template Φ_i centred at the lag, each output $y_i = \sum_k m_k\, g_i[k]$ is Gaussian with variance $\sigma_n^2\|g_i\|^2 = \sigma_n^2$, and the two outputs are uncorrelated because the templates are orthogonal.

Hence $\mathbf{y}/\sigma_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2)$, $\Lambda = \|\mathbf{y}\|^2/\sigma_n^2$ follows a χ^2 law with two degrees of freedom, and $P(\Lambda > \eta) = e^{-\eta/2}$. Maximising the Gaussian likelihood ratio over the two unknown amplitudes gives the log-ratio $\|\mathbf{P}_S\mathbf{m}\|^2/(2\sigma_n^2)$, where $\mathbf{P}_S$ projects onto the two templates. For orthonormal templates $\|\mathbf{P}_S\mathbf{m}\|^2 = \|\mathbf{y}\|^2$, so (15) is the GLRT [7, 8].

B.3 First-Order Displacement of the Energy Peak

Template truncation. Substituting $u = \tau - t$ in (16) gives $y_j(t) = a \sum_i w_i\, c_{ij}^K(t)$, with $\mathbf{w} = (\cos\beta, \sin\beta)$ and

$$c_{ij}^K(t) = \int_{-K}^{K} \Phi_i(u+t)\, \Phi_j(u)\, \mathrm{d}u. \tag{40}$$

Expanding $\|\mathbf{y}\|^2 = a^2\, \mathbf{w}^\top \mathbf{C}^K \mathbf{C}^{K\top} \mathbf{w}$,

$$\begin{gathered} \frac{\|\mathbf{y}(t)\|^2}{a^2} = \cos^2\beta\, D_1(t) + \sin^2\beta\, D_2(t) + \sin 2\beta\, X(t), \\ D_1 = c_{11}^2 + c_{12}^2, \qquad D_2 = c_{21}^2 + c_{22}^2, \qquad X = c_{11}c_{21} + c_{12}c_{22}. \end{gathered} \tag{41}$$

At $t = 0$ the off-diagonal terms vanish, because their integrands are odd on a symmetric interval, and the diagonal terms are the truncated energies (17), obtained with $\int \mathrm{d}u/(1+u^2)^2 = \frac{1}{2}\,(u/(1+u^2) + \arctan u)$. Hence $X(0) = 0$. The derivatives at $t = 0$ are the integrals of $\Phi_i'\Phi_j$ over $[-K, K]$. The diagonal ones vanish because $\Phi_i'\Phi_i$ is odd, so D_1 and D_2 are stationary at $t = 0$, and

$$\begin{aligned} X'(0) &= E_1\, c_{21}^{K\prime}(0) + E_2\, c_{12}^{K\prime}(0) = E_1 I(K) + E_2\,(2\Phi_1(K)\Phi_2(K) - I(K)) \\ &= I(K)\,(E_1 - E_2) + 2E_2\,\Phi_1(K)\,\Phi_2(K), \end{aligned} \tag{42}$$

where $I(K)$ is the integral of $\Phi_1\Phi_2'$ over $[-K, K]$. The second step integrates by parts and uses the oddness of $\Phi_1\Phi_2$. Now $\Phi_1\Phi_2' = (2/\pi)(1-u^2)/(1+u^2)^3$, whose integral over the real line is $1/2$ by (34), and whose tails decay as u^{-4}, so $I(K) = 1/2 + O(K^{-3})$. Moreover $E_1 - E_2 = 4/(\pi K) + O(K^{-3})$ and $\Phi_1(K)\Phi_2(K) = O(K^{-3})$, therefore

$$X'(0) = \frac{2}{\pi K} + O(K^{-3}). \tag{43}$$

Near $t = 0$ the phase-independent part $D = \cos^2\beta\, D_1 + \sin^2\beta\, D_2$ behaves as $D(0) + D''(0)\, t^2/2$. With full support $D_1 = D_2 = 4/(4+t^2)$, whose second derivative at $t = 0$ is $-1/2$, and truncation changes it by $O(1/K)$. Maximising $D(0) + D''(0)\, t^2/2 + \sin 2\beta\, X'(0)\, t$ over t gives

$$\hat{\tau}_{\mathrm{EP}} = \sin 2\beta\, \frac{X'(0)}{|D''(0)|} = \frac{4}{\pi K}\, \sin 2\beta + O(K^{-2}), \tag{44}$$

which is (18). Evaluated exactly, the finite-K coefficient $X'(0)/|D''(0)|$ is very close to $4/(\pi K)$ from moderate pass lengths on. The expansion neglects the terms of the energy that are cubic in t, which is why the exact displacement exceeds the first-order value on short passes (Figure 3b).

Record truncation. Here $\tilde{c}_{ij}(t)$ is the integral of $\Phi_i(\tau)\,\Phi_j(\tau - t)$ over $|\tau| \le K$, so its derivative at $t = 0$ is minus the integral of $\Phi_i \Phi_j'$ over $[-K, K]$. The same steps give

$$\tilde{X}'(0) = I(K)\,(E_1 - E_2) - 2E_1\,\Phi_1(K)\,\Phi_2(K), \tag{45}$$

with the same leading term $2/(\pi K)$ and therefore the same asymptotic law.

C Bearing and Standoff From One Crossing

C.1 Same-Shape Ratios

Let two channels share the waveform $s(t)$ up to a factor r, $m_A = r\,s + c_A + n_A$ and $m_B = s + c_B + n_B$, with constants c_A, c_B and noises n_A, n_B. Fitting a template g with a constant column by least squares gives the coefficient

$$\hat{a}(m) = \frac{\langle m - \bar{m},\, g - \bar{g} \rangle}{\|g - \bar{g}\|^2}, \tag{46}$$

where the bar denotes the mean over the samples used and $\langle \cdot, \cdot \rangle$ the sum over those samples. $\hat{a}$ is linear in m and blind to constants, so without noise $\hat{a}(m_A)/\hat{a}(m_B) = r$ for every template with $\langle s - \bar{s},\, g - \bar{g} \rangle \neq 0$. The result does not depend on the scale or the position of g, nor on which samples exist, so it holds for every truncation of the record.

Noise changes only the variance. Under white noise of variance σ^2, with $\tilde{s} = s - \bar{s}$ and $\tilde{g} = g - \bar{g}$, the SNR of $\hat{a}(m_B)$ is $\langle \tilde{s}, \tilde{g} \rangle^2 / (\sigma^2 \|\tilde{g}\|^2)$. The Cauchy–Schwarz inequality bounds it by $\|\tilde{s}\|^2/\sigma^2$, with equality when $\tilde{g}$ is proportional to $\tilde{s}$. The matched template is therefore the best choice, and a mismatched one costs SNR but adds no bias.

With a zero-mean template, $\hat{a}(m) = \langle m, \tilde{g} \rangle / \|\tilde{g}\|^2$, so the matched-filter output $\langle m, \tilde{g} \rangle / \|\tilde{g}\|$ differs from the least-squares coefficient only by the factor $\|\tilde{g}\|$, which is common to the two channels and cancels in the ratio. This is why RAT and GRAT can use the matched-filter outputs directly. Without the zero-mean step an offset c adds $c \sum_k g_k$ to the output, and since the bell Φ_1 has a large sum, an offset of a few nanoteslas is enough to bias the ratio noticeably. Absolute coefficients, as in AMP, still need the least-squares fit, because Φ_1 and Φ_2 are not orthogonal on a truncated window.

For RAT, $B_E = \sin(\varphi + 90^\circ)\,B_n$ and $B_N = \cos(\varphi + 90^\circ)\,B_n$, so $\operatorname{atan2}(Q_E, Q_N) = \varphi + 90^\circ$, modulo 180° when the common factor is negative. Summing the outputs of the two sensors keeps the ratio, because the factors $\sin(\varphi + 90^\circ)$ and $\cos(\varphi + 90^\circ)$ are common to both. For GRAT the same holds with ΔB_n in place of B_n, since the horizontal gradiometer vector lies along the normal. The ratio stops being exact when one channel contains a term that the other does not share, such as the leakage of an attitude error into the vector components, and this is what limits RAT.

C.2 Amplitude With Known Current

Coefficients. From (5), $A_2 = \sqrt{\pi/2}\,K_p \sin\delta$ comes from the vertical part of $\hat{\mathbf{T}}$ and the vertical cable field, neither of which depends on the cable direction. The vertical component is $B_z = K_p\,\tau/(1+\tau^2) = \sqrt{\pi/2}\,K_p\,\Phi_2$, so its Φ_2 coefficient is $A_z = \sqrt{\pi/2}\,K_p$. With $K_p = \mu_0 I/(2\pi d)$

both give (28). The inclination δ is taken from a geomagnetic reference model [2]. A_1 is not used, since it depends on ψ, hence on the bearing, and as the coefficient of the even Φ_1 it is the one most confounded with slow background variations on a short record.

Fixed point. Let $\lambda = d_j/d$ be the ratio of the template scale to the true standoff. On an unbounded record the columns $\Phi_1(\tau/\lambda)$, $\Phi_2(\tau/\lambda)$ and 1 decouple, $\Phi_1(\tau)$ does not project onto the odd $\Phi_2(\tau/\lambda)$, and the Φ_2 coefficient is

$$\hat{A}(\lambda) = A\,\frac{\langle \Phi_2(\tau), \Phi_2(\tau/\lambda)\rangle}{\langle \Phi_2(\tau/\lambda), \Phi_2(\tau/\lambda)\rangle} = \frac{2A}{1+\lambda}, \tag{47}$$

using $\langle \Phi_2(\tau/\lambda), \Phi_2(\tau/\lambda)\rangle = \lambda$ and $\int \tau^2/[(1+\tau^2)(\lambda^2+\tau^2)]\,\mathrm{d}\tau = \pi/(1+\lambda)$, obtained by partial fractions. Substituting into (29) gives

$$d_{j+1} = d\,\frac{1+\lambda}{2} = \frac{d+d_j}{2}, \tag{48}$$

so the iteration converges to d and each step halves the error. On a truncated record (47) no longer holds, but d remains a fixed point: at $\lambda = 1$ the template matches the record exactly on whatever samples exist, so without noise $\hat{A}(1) = A$ for every truncation. Numerically, the ratio of successive errors approaches $1/2$ on long passes and is much smaller on the short passes of a snake leg, including asymmetric ones. The starting value only changes the number of iterations.

Bearing error. Let a sample lie at the track distance s from the crossing. Then $(\mathbf{p}_s - \mathbf{p}_c)\cdot\hat{\mathbf{n}} = s\sin\theta = x$ and $(\mathbf{p}_s - \mathbf{p}_c)\cdot\hat{\mathbf{u}} = x\cot\theta$, with $\hat{\mathbf{u}}$ the unit vector along the cable. A bearing error $\Delta\varphi$ changes $\hat{\mathbf{n}}$ into $\hat{\mathbf{n}} - \Delta\varphi\,\hat{\mathbf{u}}$, so the computed coordinate is $x' = x(1-\epsilon)$ with $\epsilon = \Delta\varphi\cot\theta$. A fit of the profile shape in x' would return the relative standoff error $-\epsilon$. In AMP a template at scale d_j in x' has the scale $d_j/(1-\epsilon)$ in x. Let κ be the error ratio of the iteration, so that $|\hat{A}(\lambda)| \simeq |A|\,[1-\kappa(\lambda-1)]$ near $\lambda = 1$. Linearising (29) gives the fixed point

$$\frac{\hat{d}-d}{d} \simeq \frac{\kappa}{1-\kappa}\,\epsilon. \tag{49}$$

On a long pass, where $\kappa = 1/2$, AMP and a shape fit have errors of the same magnitude. On a snake leg κ is small and AMP keeps only a small part of the shape-fit error.

Other sensitivities. The relative error of $\hat{d}$ is the sum of the relative errors of I and of $\hat{A}$, and the error of the inclination is negligible for a reference-model value. Heave changes d itself. A fix error mixes part of A_1 into the Φ_2 coefficient of the total field, but the mixing is weak because $|A_1/A_2| = |\cos\psi|\cot\delta$ is small at high magnetic inclination, and for B_z it only enters at second order. For B_z, a linear drift of the attitude leakage over the leg is odd and projects partly onto Φ_2, which is how the attitude error enters AMP when it uses the vector channel.